2019 EXAMINATIONS

ECON 102 PRINCIPLES OF ECONOMICS **(3 hours)**

Candidates should answer ALL questions from Section A, ONE question from Section B and ONE question from Section C.
Non-programmable calculators are allowed in this exam.

Please turn over

Section A (40 marks). Each question carries equal weight.
Identify the correct answers on the answer sheet provided.

1. Other things remaining equal, which of the following is most likely to shift the demand for public transportation to the right?

(a) a decrease in the price of bus ticket
(b) a decrease in population
(c) an increase in the price of bus ticket
(d) an increase in the price for automobiles

2. Other things remaining equal, if government levies a tax on the firms producing beer, the demand curve for beer will most likely ____.

(a) shift leftward.
(b) shift rightward.
(c) remain unchanged.
(d) None of the above.

3. If ramen noodles are an inferior good then, following an increase in the price of ramen noodles, the income effect leads the consumer to consume ____ of the good and the substitution effect leads the consumer to consume _____ of the good.

(a) less; less
(b) less; more
(c) more; less
(d) more; more

4. Suppose the demand curve in a market is given by: $P = 37 – 3Q$, and the supply curve is given by $P = 2+2Q$, where P is the market price, Q is the market quantity. In equilibrium, the market price is _____.

(a) 7
(b) 14
(c) 16
(d) 21

5. Refer to the demand and supply curves given in question 4. In equilibrium, consumer surplus is

(a) less than 70.
(b) between 70 and 80.
(c) between 80 and 160.
(d) greater than 160.

Please turn over

6. Refer to the demand and supply curves given in question 4. If a per-unit tax of £1 is levied on the sellers, the consumer surplus will

(a) Increase
(b) Decrease
(c) Remain unchanged
(d) None of the above

7. Refer to the demand and supply curves given in question 4. At equilibrium, the price elasticity of demand is

(a) Less than 1
(b) Exactly 1
(c) Greater than 1
(d) None of above

8. The manufacturing process of a good causes significant negative externality. If the government levies a tax to correct the externality, the equilibrium price will ____, and the equilibrium quantity will ____.

(a) increase; decrease
(b) decrease; increase
(c) increase; increase
(d) decrease; decrease

9. Lily has allocated £15 toward coffee and tea and feels that coffee and tea are perfect substitutes. Due to differences in caffeine levels, her Marginal Rate of Substitution between tea and coffee equals 2. If coffee and tea sell for the same price, Lily will

(a) spend all £15 on tea.
(b) spend all £15 on coffee.
(c) spend £10 on coffee and £5 on tea.
(d) be indifferent between any bundle of coffee and tea costing £15.

10. In the long run

(a) at least one of the firm's outputs is fixed and at least one is variable
(b) at least one of the firm's inputs is fixed and at least one is variable
(c) all inputs are variable
(d) all outputs are variable

Please turn over

11. Consider the production function $Y = K^{½}L^{½}$ where Y denotes output, K is capital stock and L is labour input. The 24-unit isoquant passes through the point where

(a) K=16 and L=16
(b) K=16 and L=36
(c) K=9 and L=25
(d) None of the above

12. The average cost of a firm in a perfectly competitive industry is given by $q^2-10q+50$ for all values of q greater than zero. In long run equilibrium, output will be

(a) 3
(b) 4
(c) 5
(d) none of the above

13. Consider a monopolist facing a linear demand curve. At output levels higher than that at which the own price elasticity of demand is one:

(a) marginal revenue is negative
(b) marginal revenue is positive but lower than average revenue
(c) marginal revenue equals the own price elasticity of demand
(d) marginal revenue is positive so long as the good is not inferior

14. If the average cost curve is downward sloping from left to right, then
(a) marginal cost must be lower than average cost
(b) marginal cost must be higher than average cost
(c) marginal cost must be equal to average cost
(d) none of the above

Please turn over

15. By engaging in price discrimination, a profit-maximising monopolist

(a) increases its profit and reduces consumer surplus
(b) increases its costs and reduces economic welfare
(c) increases its revenue and reduces economic welfare
(d) reduces its profit and reduces economic welfare

16. Under monopolistic competition, the firm's demand curve

(a) is perfectly elastic
(b) is downward sloping from left to right
(c) is perfectly inelastic
(d) none of the above

17. In Cournot-Nash equilibrium

(a) each player recognises that no unique solution exists
(b) each player maximises the joint payoff of all players in the game
(c) there must be a prisoners' dilemma
(d) each player maximises their own payoff given other players' actions

18. In the ultimatum game, the subgame perfect equilibrium occurs where

(a) the proposer and responder determine their actions simultaneously
(b) the proposer offers the responder the least possible amount, and the responder accepts
(c) the proposer offers the responder the greatest possible amount, and the responder accepts
(d) the proposer offers an even split, and the responder accepts

19. In setting the level of employment, a firm

(a) equates the marginal cost of labour to the marginal revenue product
(b) sets the wage equal to the marginal revenue
(c) equates the average cost of labour to the marginal revenue product
(d) none of the above

20. Resources in Azerbaijan may be used either to produce 10 units of aspic or 5 units of beef. Resources in Bahrain may be used either to produce 6 units of aspic or 4 units of beef. It follows that

(a) Bahrain has a comparative advantage in aspic
(b) Bahrain has a comparative advantage in beef
(c) Bahrain has an absolute advantage in aspic
(d) Bahrain has an absolute advantage in beef

21. Macroland has a population of 50 people and 40 of them worked last year with a total output of €200,000. The average labour productivity of Macroland equalled:

(a) 200,000
(b) 4,000
(c) 40,000
(d) 5,000

22. Supporters of structural policy argue that economic growth can be stimulated and living standards improved if:

(a) the underlying structure, or institutions, of a nation's economy are changed.
(b) the money supply is controlled by the central bank.
(c) the federal budget is balanced.
(d) normative analysis is used to determine macroeconomic policy.

23. Bob's Barber Shop cut 3,000 heads of hair in the year 2017 and 3,100 in the year 2018. The price of a haircut was €7 in 2017 and €8 in 2018. If the year 2017 is the base year, what was Bob's contribution (in Euros) to real GDP in the year 2018?

(a) 21,000
(b) 21,700
(c) 24,000
(d) 24,800

24. GDP excludes important factors that affect people's well-being, such as the value of:

(a) leisure time.
(b) services purchased by households.
(c) capital goods purchased by firms.
(d) government purchases of goods and services.

25. Deflating a nominal quantity is the process of dividing a ____ quantity by a _____ in order to express the quantity in _____ terms.

(a) nominal; real quantity; nominal
(b) nominal; price index; real
(c) nominal; nominal quantity; real
(d) real; price index; real

26. If the CPI equaled 1.00 in 2000 and 1.10 in 2018 and a typical household's income equaled €35,000 in 2000 and €38,500 in 2018, then between 2000 and 2018, real household income

(a) increased.
(b) decreased.
(c) can not be determined.
(d) was constant.

27. The demand for labour in a certain industry is ND = 300 - w, where ND is the number of workers employers want to hire and w is the real wage measured in euro per day. The supply of labour in the same industry is NS = 200 + w, where NS is the number of people willing to work. If the minimum wage is set at €60 per day, how many workers will be unemployed?

(a) 10.
(b) 20.
(c) 30.
(d) 40.

28. Data for an economy show that the unemployment rate is 10%, the participation rate is 80% percent, and 200 million people 16 years or older are not in the labour force. How many people are in the labour force in this economy?

(a) 80 million.
(b) 800 million.
(c) 200 million.
(d) 1,000 million.

29. If an economy maintains a small rate of growth for a long period of time, then the size of the economy

(a) can only increase by a small amount.
(b) will stay nearly constant.
(c) can increase by a large amount.
(d) will shrink.

Please turn over

30. Alpha has €40,000 of capital per worker, while Beta has €5,000 of capital per worker. According to the principle of diminishing returns to capital, an additional unit of capital will increase output ____ in Alpha compared to Beta, holding other factors constant.

(a) more
(b) less
(c) by the same amount.
(d) either more or less.

31. The investment demand curve indicates that there is a(n)

(a) inverse relationship between the real interest rates and the level of investment spending, all other things equal.
(b) positive relationship between the real interest rates and the level of investment spending, all other things equal.
(c) direct relationship between the real interest rates and the level of investment spending, all other things equal.
(d) positive relationship between the determinants of investment and the level of investment spending, holding interest rates constant.

32. If planned aggregate expenditure (PAE) in an economy equals 2,000+0.8Y and potential output (Y*) equals 11,000, then this economy has

(a) an expansionary gap.
(b) a recessionary gap.
(c) no output gap.
(d) no induced expendutres.

33. Denoting income and interest by Y and i respectively, if taxes increase, the IS-LM equilibrium will_____

(a) yield a lower Y and i.
(b) yield a lower Y but a higher i.
(c) yield a higher Y but a lower i.
(d) yield a higher Y and i.

34. With liquidity preference theory, when money is held for 'speculative purposes', the speculation is that

(a) interest rates will rise
(b) commodity prices will rise
(c) bond prices will rise
(d) import prices will rise

Please turn over

35. When it engages in ‘quantitative easing’, a central bank

(a) buys long-term government bonds
(b) sells long-term government bonds
(c) sells short-term government bonds
(d) purchases short-term government bonds

36. A yield curve for bonds shows

(a) the return on investments overseas
(b) the profitability of long-term investments
(c) rates of return across a range of time-to-maturity
(d) a range of discount rates offered by a central bank

37. With the original Phillips curve, the variable shown on the vertical axis is the proportionate rate of change in

(a) real wages
(b) prices
(c) productivity
(d) nominal wages

38. Indicate the missing words. As a _ (i) _ bank makes open-market purchases of bonds, it spends newly created _ (ii) _ money. As bond prices then rise, bond yields _ (iii) _. With competition in financial markets, interest rates fall generally. In short, as the money _ (iv) _ rises, interest rates fall.

(a) (i) central (ii) broad (ii) fall (iv) supply
(b) (i) commercial (ii) broad (ii) fall (iv) supply
(c) (i) central (ii) base (ii) fall (iv) supply
(d) (i) central (ii) base (ii) rise (iv) supply

39. By dropping the assumption that the _____ is constant, the Aggregate Demand schedule can be derived from the ISLM model

(a) growth rate
(b) general level of prices
(c) unemployment rate
(d) level of taxation

40. A yield curve for bonds shows

(a) the return on investments overseas
(b) the profitability of long-term investments
(c) rates of return across a range of time-to-maturity
(d) a range of discount rates offered by a central bank

Please turn over

Section B (30 marks) Answer ONE question

21. (a) Explain what is meant by "externalities". Provide two examples to illustrate the idea and why externalities could lead to inefficiency. (10 marks)

(b) Explain the concept of the "Tragedy of the Commons" and provide two ways to mitigate the problem. (10 marks)

(c) Use examples to illustrate the two types of problems that arises when asymmetric information exists. (10 marks)

42. What assumptions underpin the kinked demand curve theory of oligopoly? Using an appropriate diagram, consider the implications of this model for the stability of prices in an oligopoly in which marginal costs flucuate over time.

43. (a) Determine the values of x and z that maximise the function
$y = 40x - \frac{5}{2}x^2 + 2xz - 2z^2$ (15 marks)

(b) A firm can sell its products in two countries. Quantity demanded in the first country is given by

$q = \frac{99}{2} - \frac{1}{2}p$

(where p denotes price in the first country), and in the second country

$Q = 99 - P$

where P denotes price in the second country.

The firm's total output is q+Q which it can produce at a total cost (TC) of

$$TC = 78q + 78Q + qQ + 0.5q^2 + 0.5Q^2$$

Find
(i) the prices and outputs in each country that maximise profits (9 marks)
(ii) the price elasticities of demand in each country at the profit maximising prices (6 marks)

Please turn over

Section C (30 marks) Answer ONE question

44. Consider the following economy:

C=360-200r+0.2Y
I=120-200r
G=120
(M/P)D=100+0.2Y-200r
P=2
Ms=300

where C=consumption, Y=output, I=investment, G=government spending, r=interest rate, (M/P)D =real demand for money, P=prices and Ms =money supply.

a. Find the algebraic expression for the IS curve and the LM curve and explain why the slope of the IS curve is negative while the slope of the LM curve is positive. (10 marks)
b. Find the equilibrium interest rate (r) and the equilibrium level of output (Y). (10 marks)
c. Suppose the central bank increases the money supply (Ms) from 300 to 400. What is the new equilibrium level of interest rate and output? Explain your answer using words and a relevant graph. (10 marks)

45. Suppose the economy is at its potential output *Y** and experiences a sudden increase in net exports. Use an IS-LM diagram to illustrate how the central bank might prevent the emergence of an expansionary gap. (30 marks)

Please turn over

46. Examine the mechanism of *fractional reserve banking* and the associated concept of the *money multiplier*. (30 marks)

End of paper